NATIONAL UNIVERSITY OF MODERN LANGUAGE ISLAMABAD

Assignment 03

Submitted to:

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Submitted by:

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Roll No: …………………… SP21379

Subject: …………………… information security

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Answer No 01

1. **Vulnerability of IEEE Wireless Security**
   IEEE 802.11 (Wi-Fi) standards have historically suffered from security vulnerabilities due to weak encryption, flawed protocols, or implementation errors. A notable example is the WEP (Wired Equivalent Privacy) vulnerability.

   WEP Vulnerability:
   WEP uses a static 40-bit or 104-bit encryption key, which is easily crackable due to weak initialization vectors (IVs). Attackers can capture enough data packets to perform statistical attacks (e.g., using tools like Aircrack-ng) and decrypt the key within minutes.

   Example Attack:
   An attacker eavesdrops on a Wi-Fi network using WEP. By capturing thousands of IVs, they exploit the key reuse weakness to recover the password, gaining unauthorized access to the network and potentially stealing sensitive data.

   Mitigation:
   Modern Wi-Fi networks use WPA3 (or WPA2 with AES), which replaces WEP with stronger encryption and dynamic keys.

Answer No 02

2. **Cybercrime: Real-Life Scenarios & Prevention**
   Definition:
   Cybercrime refers to illegal activities conducted via digital means, such as hacking, phishing, ransomware, or identity theft.

   Real-Life Scenarios:
   Phishing Attack (2020 Twitter Bitcoin Scam):
   Hackers compromised high-profile Twitter accounts (e.g., Elon Musk, Barack Obama) to post fake Bitcoin giveaway links, stealing over $100,000.
   Cause:
   Social engineering and employee credential theft.
   Ransomware (2021 Colonial Pipeline Attack):
   Hackers encrypted the pipeline's systems, demanding $4.4 million in Bitcoin, causing fuel shortages in the U.S.
   Cause:
   Exploited an unused VPN account with weak password security.
   Identity Theft (2017 Equifax Breach):
   Attackers stole personal data (SSNs, addresses) of 147 million people due to unpatched software vulnerabilities.
   My Role in Stopping Cybercrime:

As an Individual:
Use strong, unique passwords and enable MFA (Multi-Factor Authentication).
Avoid clicking suspicious links or sharing personal data online.
Keep software updated to patch vulnerabilities.
As a Professional (if in IT):
Educate others about cybersecurity best practices.
Implement network security measures (firewalls, encryption).
Report cybercrimes to authorities (e.g., FBI’s IC3 in the U.S.).